

The Role of Machine Learning in Networks and Network Security

Machine learning (ML) focuses on the development of computer programs that can access data and use it for further learning. In this era of automation, due to the great success of artificial intelligence, ML is being integrated into almost everything. In this article, we will see how ML is solving the problems of complex networks and helping with network security.



“Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.”

— Arthur Samuel, 1959

According to Brownlee J. in his work ‘Practical Machine Learning Problems’, there are four broad categories of problems that can leverage ML, namely, clustering, classification, regression and rule extraction. In clustering problems, the main objective is to group similar data and increase the gap between the groups. In classification and regression-based problems, the goal is to map a set of new input data to a set of discrete or continuous-valued outputs, respectively. Rule extraction problems are essentially

different; here, identifying statistical relationships in data is the main goal.

Machine learning for/in networking

A lot of research is being done today in adopting AI as a tool for solving the problems of modern computer networks, as these are becoming increasingly complex and dynamic. AI is revolutionising network services by making more informed decisions based on the huge data available. It is a central component in cognitive networks and communication research.

The various applications of AI/ ML/ deep learning in computer networks and communication are (but not only limited to):

- Autonomous management of data centres and cloud infrastructures
- Modern applications of AI or ML in the management of networks and services
- Cyber security, including anomaly detection, malware detection, etc
- Modern approaches in cognitive computing
- Self-managing middleware and tools for extreme scales
- Big Data analytics frameworks for networking data
- Network monitoring and performance anomaly detection
- Machine learning for multimedia networking
- Resource allocation in networks using ML
- Deep learning and reinforcement learning in network control and management
- Applications of game theory in computer networks
- Applications of evolutionary computing in network optimisation
- Applications of AI in network configuration tuning
- Testing of cyber-physical systems using AI and ML
- Autonomous sensors networks and self-organising systems
- Adaptive stream-mining and resource-efficient scientific computing